

Total No. of Questions : 4]

SEAT No. :

PD291

[Total No. of Pages : 1

[6411]-67

B.E. (E & TC Engineering) (Insem)

NANO ELECTRONICS

(2019 Pattern) (Semester-VIII) (Elective-VI) (404192 B)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) Name the tools for measuring Nano structure? Explain Transmission Electron Microscope(TEM). [8]

b) Explain Scanning Prob Microscopy (SPM) With neat Diagram. [7]

OR

Q2) a) What are the limitation of Silicon Materials. [7]

b) Explain Lithography Process Steps. [8]

Q3) a) Explain Dielectric Material for future Transistor. [8]

b) Explain Atomic Force Microscopy (AFM). [7]

OR

Q4) a) With Neat diagram Explain Nano CMOS-Devices. [7]

b) Explain Silicon nano Crystal Non Volatile Memories. [8]

